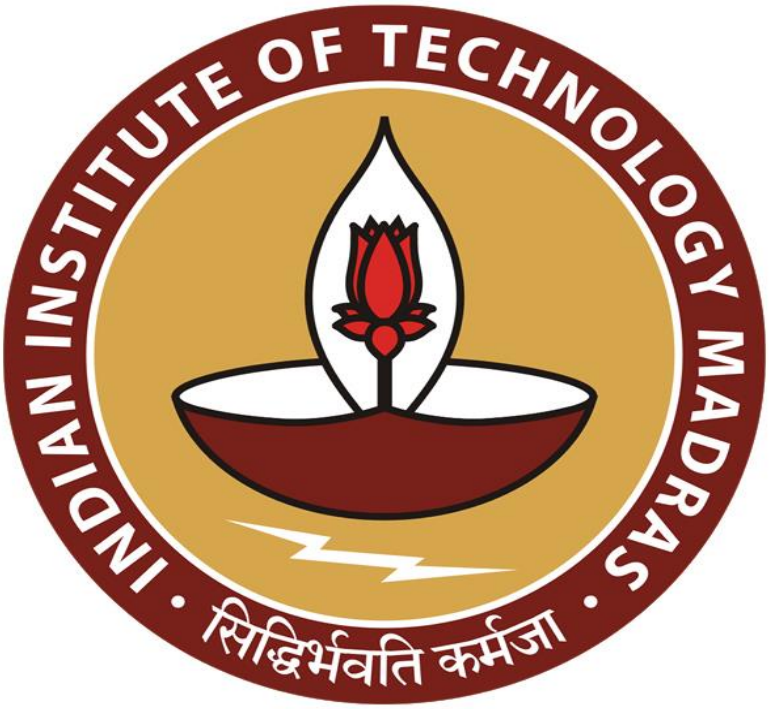
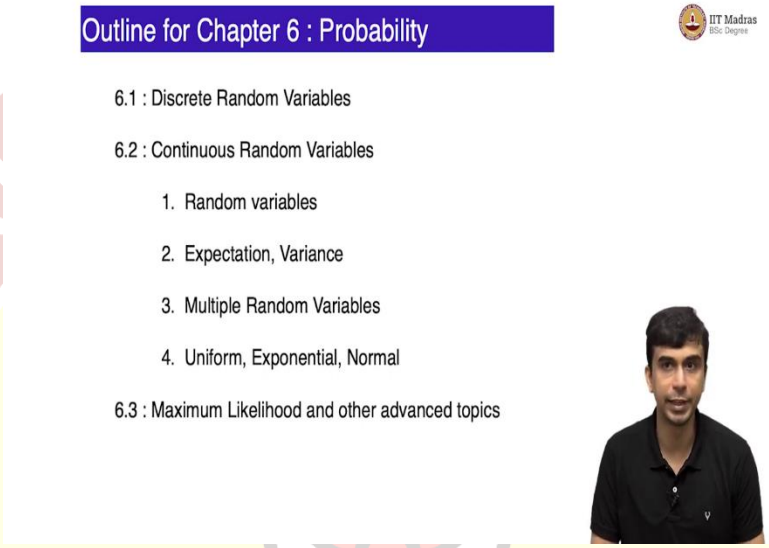


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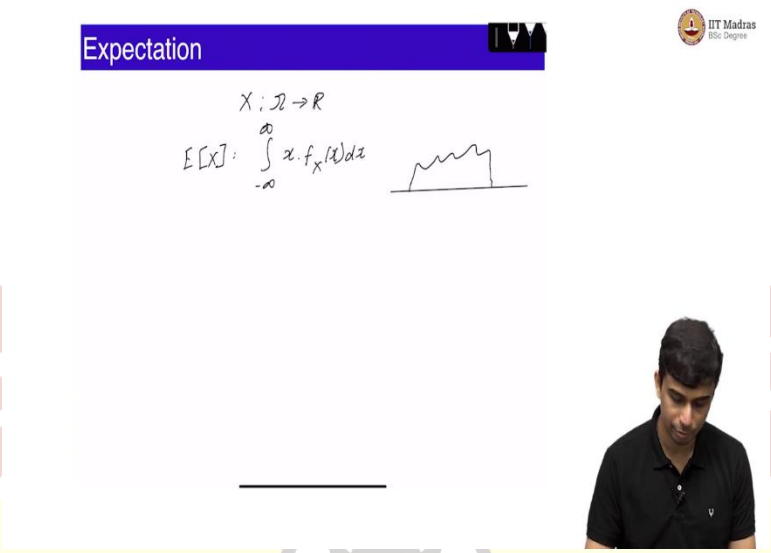
Professor Harish Guruprasad Ramaswamy
Department of Computer Sciences and Engineering
Indian Institute of Technology, Madras
Expectation

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Now we will move to the next topic here which is expectation and variance of continuous random variables. You see that varies in the same terminology which is we will see the relation to discrete random variables in fact it is the same definition. Just that in all place where you had summation they will be replaced by integration that is the only difference.

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What is the, let us say you have a continuous random variable x which is from omega to reals. What is the expectation with density fx? What is the expectation of this random variable? The formal definition is that it is integral - infinity to infinity x into $f_X(x)$ dx. Here you can see the similarity to the probability mass function which is discrete random variable analog where you had sum over $\int_{-\infty}^{\infty} x \ f_X(x) dx$.

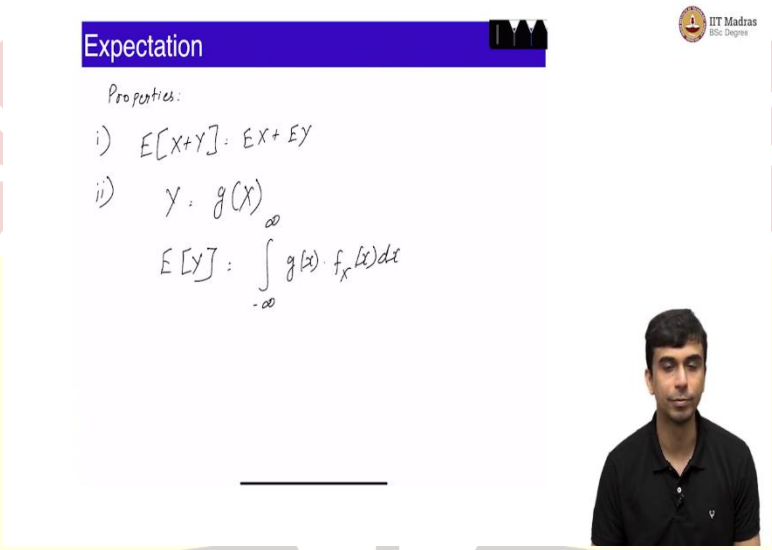
So, here you have integral - infinity to infinity x into $f_X(x)$ dx. The only difference is that you have a dx here and integration which you had just with replaces the summation for the discrete random variable case. The physical meaning remains the same where in the case of discrete random variables I asked you to imagine the probability mass function as a literal mass on a scale and the center the point at which you can balance the scale is the expectation.

And in an exact manner, exactly some analogous way you can define you can let us say you have the probability density function of this some random variable let us say it is like this. We can have a density function which looks like this curve. And the point you can imagine this as a scale where the mass of the scale at that particular length is proportional to the height of the curve. You can imagine a scale like that and the point at which you can balance that scale is exactly equal to the expected value of x.

So, the center of mass for a finite a discrete set of objects corresponds to expectation in the case of discrete random variables. But a center of mass of sorts for continuous objects that corresponds

to expectation for continuous random variables. So, that is expectation you can we will see some examples of expectation how we will compute expectation for some simple examples. Let us take the simplest. But before we do that let us see some properties of expectation. So, some of the basic properties of expectation are as follows. Let us do the properties.

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The first property is that linearity of expectation that is you have 2 random variables X and Y. Expected value of X + Y is equal to E X + E Y you have two random variables X and Y. This is one of the most basic properties of expectation X and Y can be any two random variables of course they must be from the same experiment. You cannot have 2 different experiments and compare and variables they have to be the same experiment.

But they can be the same random variables they can be completely independent they can be however, the case may be it is linearity of expectation is one of the most basic properties of expectation. I am not going to define with it is pretty easy to prove this but I am not going to prove that anyway. Then there is the other property which is let us have let us say Y is equal to some function of the random variable X see if Y is equal to g(x).

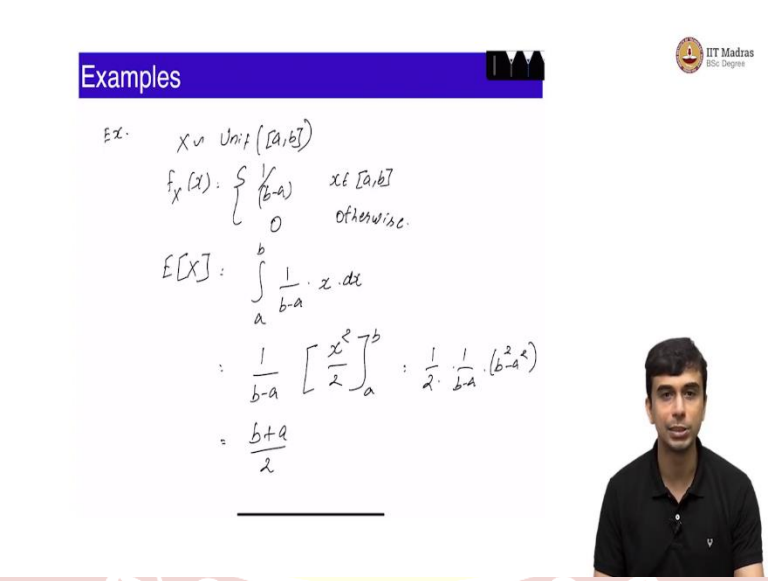
What is the expected value of Y? Of course, you could compute the probability density function of Y. And then you could apply this from first principles you can compute the expectation of Y. But there is a much easier expression which is analogous to the discrete random variable case EY

is simply equal to $\int_{-\infty}^{\infty} g(x)\, f_X(x)dx$. You can see this much simpler because I have not computed the density function of Y at all.

I have just used the density function of x but instead of multiplying by x I am multiplying by g of x. What is the logic behind this? It is a pretty intuitive logic it is pretty easy to see for discrete random variables. A very similar logic works for continuous random variables also. But I am not going to prove that. For discrete random variables you can recall the logic you can go back and check the logic if you want.

But for continuous random variables it is exactly the same logic where you do not need to compute the density function of Y. If you have the density function of X and Y is g(x). To compute expectation of Y you can just have $\int_{-\infty}^{\infty} g(x)\, f_X(x)dx$. So, this is a pretty powerful shortcut. And you can use this for computing expectation of more complex random variables.

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So, there are several other properties of expectation but I am not going to bother with those. These two the two properties that I mentioned are perhaps the most important properties of expectation. So, let us go ahead with some examples. Let us take the simplest example. Which is, X is distributed as let us say a uniform is in the range of a to b what do I mean by that? Well, fx or by that I mean $f_X(x)$ is equal to 1 by b - a if x is in the range of a to b.

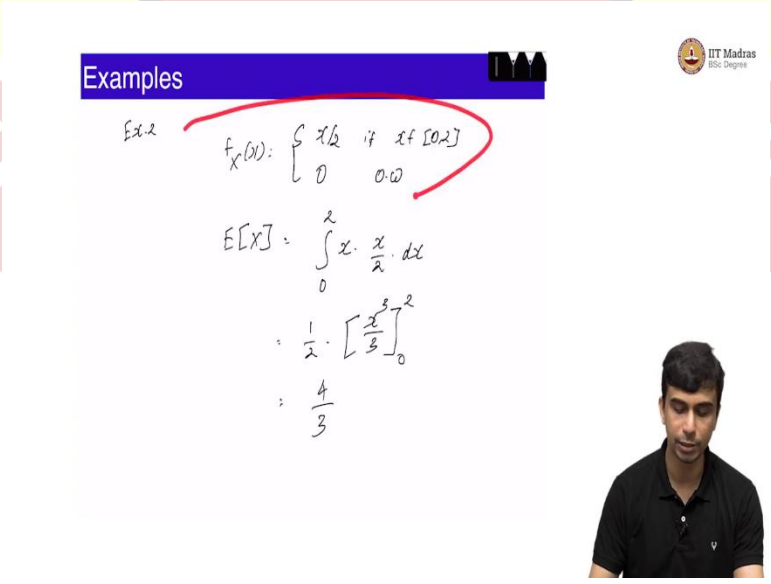
And 0 otherwise. Can you easily see that this is a valid probability density function because the if you drew the density of if you could graph the density function you would look like a rectangle and the area of rectangle is exactly equal to $\frac{1}{b-a} \times (b-a)$ which is 1. Where we will assume that b is greater than a. This is the density function of the uniform. This is a uniform distribution, uniform density function.

What is the expected value of this random variable? Well, you have integral - infinity to infinity. But for all purposes it is only a to b that matters because outside a to b it is going to be equal to 0. So, it is going to $\int_b^a \frac{1}{b-1}\, x\, dx$. This is the $f_X(x)$ this is x this is dx. This is the three terms. $\frac{1}{b-a}$ is a constant you can take that out.

x dx is simply $\frac{x^2}{2}$ in the limit a to b. Which is equal to half into $\frac{1}{2}\frac{1}{b-a}$ *(b^2-a^2). And what is that? Well, (b^2-a^2). can be factorized as (b + a) *(b –a) this is equal to (b + a) by 2. So, this is not particularly surprising.

If X is uniform, the range of a to b expected value of X is clearly the average of the 2 end points mean if you have a uniformly distributed rod between a and b you can balance it exactly at the point a + b by 2 which is the midpoint of a and b. Which is not a surprising thing. But it is good to check do a sanity check of our definitions?

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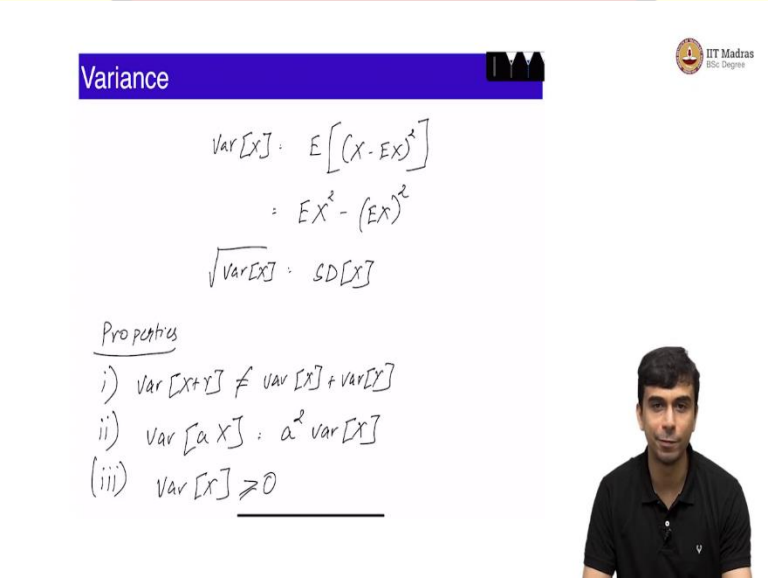


Let us do maybe a slightly more complicated example. Let us take x to be the right angle triangle example that we had$f_X(x)$ is equal to x by 2 if x belongs to 0 to 2, 0 otherwise. So, this is also we check this it is a valid density function, what is the expected value of this random variable? Once again you do not need to integrate from - infinity to infinity you can just integrate from 0 to 2, integral 0 to 2 x into $f_X(x)$ dx x into x by sorry x into x by 2 dx.

So, this is going to be x squared by 2, this is half x cubed by 3 is in the range 0 to 2. So, is 1 by 2 into 8 by 3 which is 4 by 3. So, what this says is the expected value of x is 4 by 3. And this kind of does make sense because $f_X(x)$ at x takes values in the range of 0 to 2 but it is much more likely to take larger values than smaller values. Why is that because the density increases, as you go from 0 to 2.

The density is not constant between 0 to 2, if the density was constant between 0 to 2, the point at which you can balance the scale is exactly at the midpoint. But you have a triangular kind of plot where the density increases as you move from 0 to 2. So, x is you should immediately guess that E x should be greater than 1. But obviously, it should be less than 2 because the maximum value itself is 2. So, E x is equal to 4 by 3 is the answer here. So, these are some examples, you can just use simple first principles to get the expected value of x from the density function if you happen to have the density function.

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So, then we will move to the next object, which is the variance is also the same as we defined for discrete random variables. You can have a random variable continuous random variable X variance of the random variable x is simply the expected value of the squared deviation of X from E X. You can do a simplification as the same way that you did for discrete random variables you would get E X variance X expected value of X squared - expected value of X the whole square.

The intuitive meaning we have seen this already, in the case of discrete random variables, it essentially measures on average, how much does X deviate from its expectation? So, this average, on average, how much does X deviate from its expectation, So, there is a squared here to ensure that there is no negative numbers? You could argue, well, you can use absolute value as well. But the standard definition of variance is to use a squared deviation.

The square root of the variance of a random variable is often called the standard deviation. Square root of variance of x is the standard deviation. This is another standard as this is the definition. So, let us compute let us do some simple examples. And before we go there let us see what are the properties of the variance of X? Let us do the properties. Unlike expectation you do not have linearity of variance you do not have that variance of X + Y in general, is not equal to variance of X + variance of Y.

Of course, special cases where it does happen, we will take a look at this later on. But in general, you do not have variance of X + Y is equal to variance of X + variance Y. And you do not even have the well, you do have a (())(13:04) the variance of a times X is equal to a squared times variance of X. And say A is some constant let say 2. If you have X, the variance of X is let us say 5. What is the variance of 2 times X?

That is 4 times the variance of X, which is would be 20. So, that that is an example. And a is some constant variance of a X. Recall, a X is also a random variable. So, you can imagine Y as equal to a X, Y is also a random variable. Variance of Y is simply equal to a squared times variance of X I can guess why the a squared comes from because you have, you can essentially take a squared outside of this and the linearity of expectation, you can take a square outside.

And you will get a square times variance of X. This is pretty easy to prove. And this is also easy to check it is easy to come up with an a case where variance of X + Y is not equal to variance of

X + variance of Y. The easiest example would be Let us say X is any nonzero variance random variable Y is equal to - X. If you said Y equal to - X variance of X + Y is equal to 0.

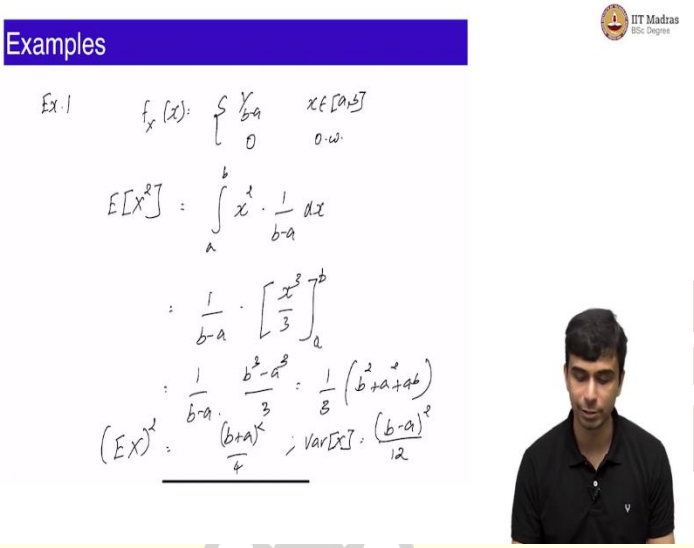
Variance of 0 means of a constant function is 0. But variance of X + variance of Y is not 0 with because variance of X is nonzero variance of Y is also nonzero. And there is another trivial property which is that variance is always positive or non negative. It can be 0 also variance of x is greater than or equal to 0. Why is that? Well, that is because the consequence of the squared expected value of X - E X the whole squared.

This is the average of the random variable given by X - E X the whole squared. If you take let us say, if you define Y to be equal to X - E X the whole squared, Y is a positive random variable. So, expectation of a positive random variable has to be positive or at least it cannot be negative. It can be 0 if the random variable happens to be 0. When is it 0? That is the next question.

So, Y let us say if you say Y equals X - E X the whole squared the only time when the average of Y is equal to 0 is if Y is always equal to 0. So, when is Y always equal to 0? That means that X is always equal to E X. That effectively means that variance of X is equal to 0. Only if if and only if X is a constant. X is always equal to the same thing regardless of what outcome you can you do have an experiment.

Let us say, let us take the height measurement, height measurement experiment. I bet I can say X is equal to X is equal to 3. If I say X equal to 3 then it does not matter what the height mean, I outcomes do depend on the experiment. But this particular random variable is a constant. So, the variance is 0. So, only constant random variables have variance to be equal to 0.

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So, let us do some simple examples. Let us take the example of uniform first. In fact, that is the only example you will do. Let us say x is distributed as uniform between a to b we have already seen this that $f_X(x)$ is equal to $\frac{1}{b-a}$ is in the range of a to b and 0 otherwise. What is the variance of x, we will use the other definition of variance of x which is $E[x^{2}] - (E\ x)^2$.

We already know what E x is that is $\frac{b+a}{2}$. What is E x squared? Let us do one. What is the expected value of X squared? Here, you can use the function of a random variable expectation rule that I mentioned earlier. That is, so you can use integral - infinity to infinity or rather a to b integral a to b x squared into $f_X(x)$ dx. What is that $f_X(x)$ dx within the range of a to b it is simply $\frac{1}{b-a}$.

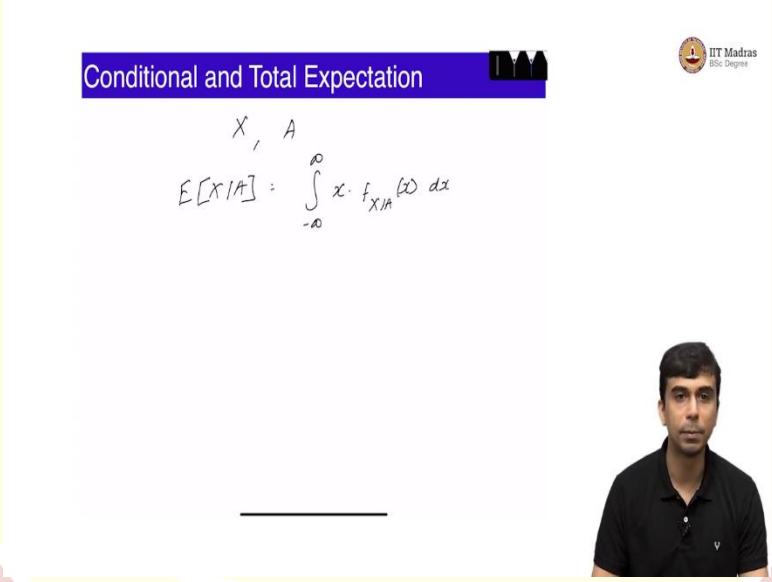
So, this is equal to $\frac{1}{b-a} \times \frac{x^3}{3}$ is in the range a to b which is $\frac{1}{b-a} \times \frac{b^3-a\text{^}3}{3}$. This is exactly equal to $\frac{1}{3} \times (b^2 + ab + b^2)$. This is expected value of X^2. But what is variance of X? Variance of X is simply the expected value of X squared - E X the whole square. What is E X squared? E X the whole square E X is (b + a)/2 E [X] the whole squared is $(b + a)^2/4$.

So, you can take this term and subtract these 2 terms 1 by subtract this term from the term $E [X^{2}] - (E [X])^2$. And you would get variance of x to be equal to $(b – a)^{2}/12$. So, this is simple algebra, you can do that and get variance of X is $(b – a)^{2}/12$. And when is variance of X equal to 0 when b is equal to a when b equals a variance of x equals 0?

And what does it mean to have b equal to a well X always is going to take the value of a which is equal to b that is the only time when you have uniform random variables having a 0 variance. So, I have mentioned variance as a measure of spread of the random variable around the mean. And this also kind of captures that. So, for uniform random variables b - a is the measure of is exactly equal to the spread of this random variable.

And the variance of X is simply $(b- a)^2/ 12$. For this particular case, in general also, this is true that is variance of random variable measures essentially the spread how if you drew the if you graph the density of density function of this. Where, what is the width of the density function. How bigger graph should you have for drawing this that is given by the variance.

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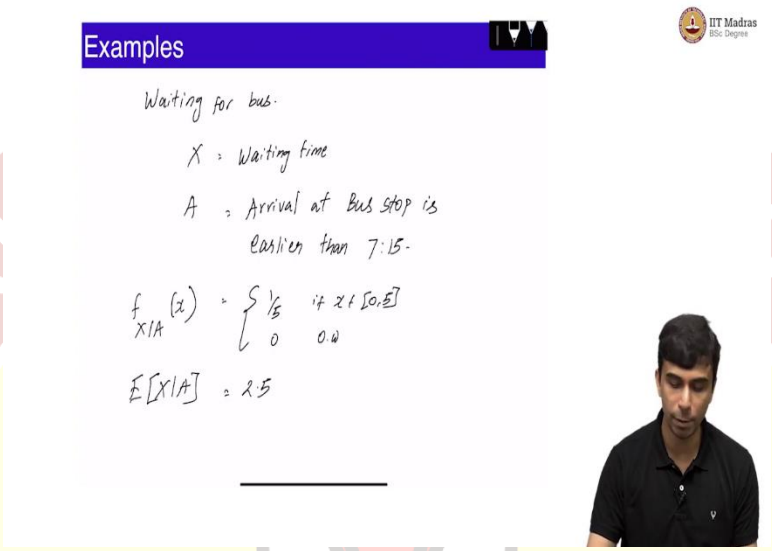


Now we will move to another related another concept, which is conditional expectation. We have already seen conditional density functions and we can immediately guess what conditional expectations are. Can have let us say X is a random variable, X is a random variable and A is an event we can have expected value of X given A is simply equal to integral - infinity to infinity x times conditional density function.

The only difference from expectation is that instead of using the standard density function you are going to use the conditional density function. That is the only difference. Condition expectations are also a very powerful tool. And in machine learning in particular condition expectations are in

fact one of the fundamental tools for building on top of on which regression classification everything is build. Let us see a simple example again for conditional expectation.

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We will go back to the example that I mentioned at the beginning. Let us say you have where you are going to a bus stop uniformly between 7:10 to 7:30. And there is the bus arrives there is a bus that arrives every 15 minutes from 7:15 7:30 and so on. How would you compute the expected let us say X is the amount of time that you are waiting for the bus? What is the expected amount of time that you are going to wait for the bus?

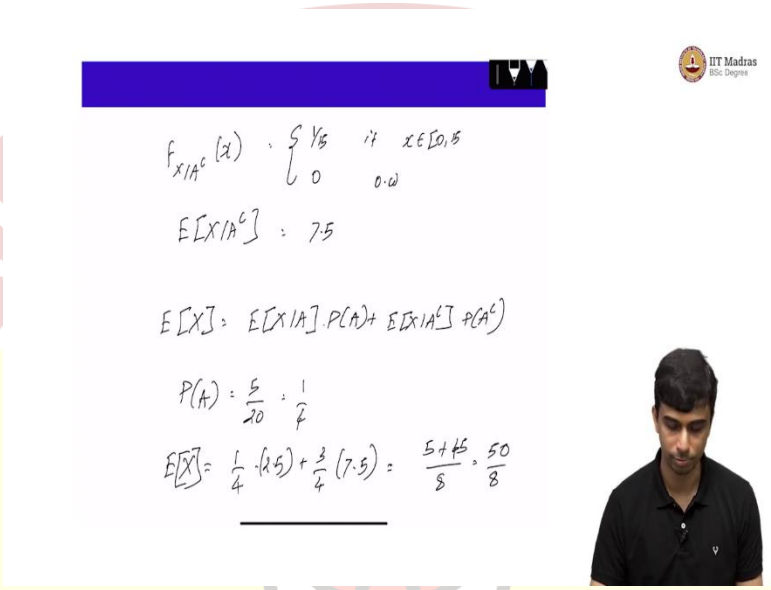
Now, you could compute the density function of this random variable that is the waiting time and directly do it. But in this case instead of computing the directly the density function of X it is actually easier to compute the density function given an event. Let us waiting for bus experiment. Let us now define X is the amount of waiting time x equal to waiting time X is the waiting time.

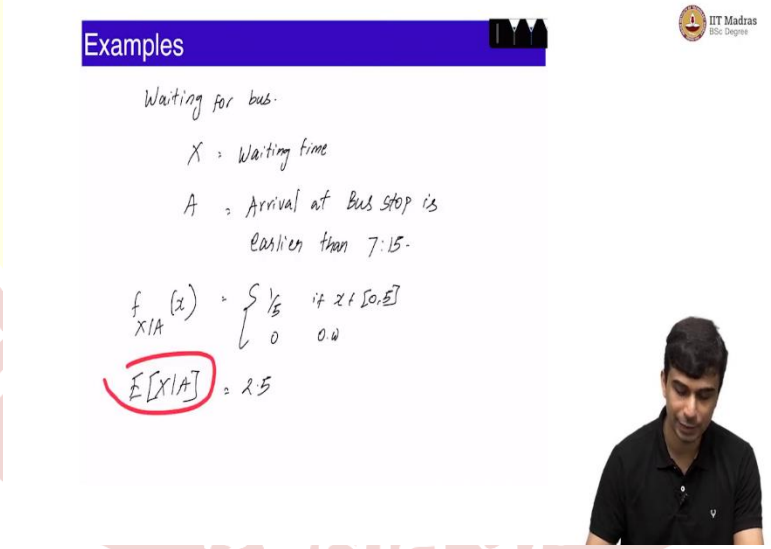
Let us define A to be the event that X is between R, A is the event that you arrive at the bus stop arrival at bus stop is earlier than 7:15. So, A is the arrival so, if you arrive given that you have arrived earlier than 7:15, what is the density of X? What is fx given A? You can by logic you can say given that you arrived earlier than 7:15. You could have arrived any point between 7:10 to 7:15. So, you can say that this is uniform in the range of 7:10 to 7:15.

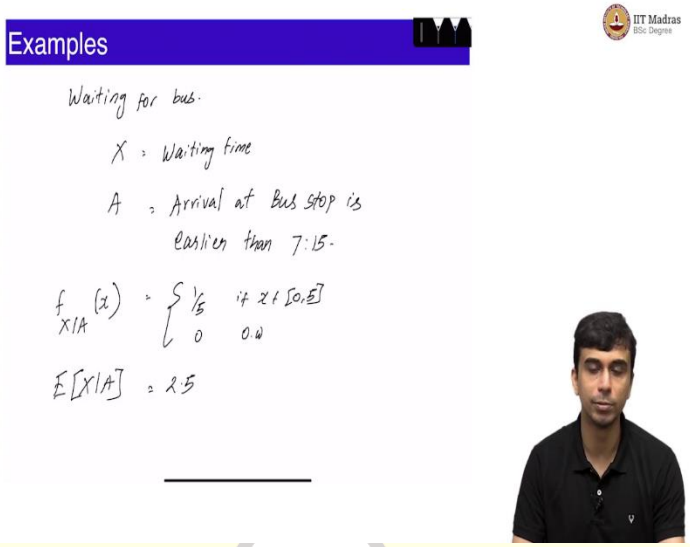
So, uniformly between the total waiting time can be any time between 0 to 5 minutes. So, that is going to be equal to 1/ 5 if x belongs to 0 to 5, 0 otherwise f(x) given A of x is this. So, conditional

expectation of X given A is also pretty easy to compute because this is exactly equal to uniform 0 to 5 which means that the expected value of X given A is equal to 2.5. That is fine. And similarly, you can do the same thing for A compliment also. That is the interesting part here.

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A compliment means that you have arrived at the bus stop later than 7:15. But do you know for sure that you will arrive before 7:30. So, the total waiting time here would be uniformly between 15 and sorry 0 to 15. You could have arrived exactly at 7:15 or you could have arrived exactly at 7:30. Which would correspond to the extremes of 15-minute waiting time and a 0-minute waiting.

So, fx given A compliment would be equal to 1/ 15 if x belongs to 0 to 15 and 0 otherwise. fx given A compliment of x is also easy to write. So, this is exactly uniform 0 to 15. And you can compute expected value of X given A compliment also pretty easily which is simply (b + a)/ 2 which is 7.5. Now, it is very easy to compute the expected value of X and given A and expected value of X given A compliment. These are very easy to compute.

But that is not what we wanted. We want the expected value of X. And here is where you can use the what is called the total probability law which is written as follows. Expected value of x is simply equal to expected value of X given A into probability of A + expected value of X given a compliment into probability of A compliment. This is called the total property law. And in this case to compute the E of X we already have E X given A.

And we also have E X given A compliment. Now we just need to compute probability of A and probably A compliment. What is probability of A? What is it is exactly equal to the probability that you arrive at the bus stop before 7:15. And we know that we know from our from our probability measure that you are uniformly between 7:10 and 7:30. So, probability of A is exactly equal to 5 by 20 which is 1 by 4.

So, now we have E X is simply equal to 1 by 4 into 2.5 + 3 by 4 into 7.5. What is that? Well, 1 by 4 into 5 by 2 + 3 by 4 into 5 by 2. Which is 8 to 5 + 45 is 50 by 8. Which is slightly more than 6 minutes so this 6.25 minutes. You could have computed f(x) the density function of X directly. I am not saying it is impossible but you can see the density function of X given A and density function of X given A compliment is significantly easier to compute.

And they happen to be nice uniform distributions. So, which means that the expectation is also easy to write. And E X becomes pretty easy to compute B using E X given A, A is given A compliment and probability of A and probably A compliment. So, total the total expectation is a pretty powerful tool for computing expectations. And it is especially in cases where you can you have a very easy nice split of outcomes.

In this case, you either catch the bus at 7:15 or 7:30. So, there was a clear 2 possible groups of outcomes and conditioning on each group makes life significantly easier. So, with that we can wrap up this lecture.